Learning a Continuous Attractor Neural Network from Real Images

Xiaolong Zou[1,*], Zilong Ji[1,*], Xiao Liu[1], Yuanyuan Mi[2], K Y Michael Wong[3], and Si Wu[1]

[1]State Key Laboratory of Cognitive Neuroscience & Learning, IDG/McGovern Institute for Brain Research, Beijing Normal University, Beijing 100875, China.
[2] Brain Science Center, Institute of Basic Medical Sciences, Beijing 100850, China.
[3] Department of Physics, Hong Kong University of Science & Technology, Hong Kong.
[*] Equal contribution.

Abstract. Continuous attractor neural networks(CANNs) have been widely used as a canonical model for neural information representation. It remains, however, unclear how the neural system acquires such a network structure in practice. In the present study, we propose a biological plausible scheme for the neural system to learn a CANN from real images. The scheme contains two key issues. One is to generate high-level representations of objects, such that the correlation between neural representations reflects the sematic relationship between objects. We adopt a deep neural network trained by a large number of natural images to achieve this goal. The other is to learn correlated memory patterns in a recurrent neural network. We adopt a modified Hebb rule, which encodes the correlation between neural representations into the connection form of the network. We carry out a number of experiments to demonstrate that when the presented images are linked by a continuous feature, the neural system learns a CANN successfully, in term of that these images are stored as a continuous family of stationary states of the network, forming a sub-manifold of low energy in the network state space.

Keywords: Continuous attractor neural network, correlated patterns, modified Hebb rule, deep neural network

1 Introduction

The brain performs computation via dynamics of neural circuits formed by a large number of neurons. A neural system can reliably retrieve the stored information even when external inputs are noisy. This can be viewed as attractor computation mathematically, that is, the network dynamics enables the neural system to reach to the same stationary state (attractor), once an external input falls into the basin of attraction. Hopfield network is such a representative attractor model, which has been applied successfully to explain associative memory [11].

The experimental and theoretical studies have shown that there exists another form of attractor model, called continuous attractor neural networks(CANNs)[6, 12–14]. CANNs have been successfully applied to describe the encoding of a number of continuous features in neural systems, such as orientation, moving direction, head direction, and spatial location of objects (see [7] and references therein). The key property of a CANN is that it holds a continuous family of stationary states, which form an *approximately* flat sub-manifold of low energy in the network state space; whereas, in the Hopfield network, attractor states are isolated with each other with high-energy barriers.

Despite the importance of CANN being widely recognized in the field [6], the mechanism of how the neural system acquires such a structure remains largely unknown. For certain features, such as orientation tuning in V1, it is known that the neural system gains a rough structure from heredity and continually refines it after birth based on visual experiences. However, for many other less directly perceivable features, in particular, for those varying with computational tasks, the neural system needs to develop a CANN in a data-dependent way. For instances, in the monkey experiments, Logothetis et al. found that after training, neurons in the inferior temporal cortex can form a CANN to encode the view angle of an object [15]; and Mante et al. found that after training, neurons in PFC can form a CANN to encode the evidence values of different sensory cues [16]. It is far from clear how the neural system develops CANNs in those on-line computations.

To learn a CANN, it faces two fundamental challenges technically. One is to generate proper neural representations. In effect, the unsupervised Hebb rule, or its variations, is to convert the overlap between neural representations (the correlation) into the strengths of neuronal connections, such that the similarity/dissimlarity between objects is encoded in the network structure. Thus, it is crucial that the overlap between neural representations reflects the sematic, rather than a superficial (e.g., different spatial locations), relationship between objects. The other challenge is to learn correlated representation patterns. Using the conventional Hebb rule, the Hopfield model can learn statistically independent memory patterns, but once the patterns are correlated, the performance of the Hopfield model is deteriorated dramatically.

In this study, we propose a biologically plausible scheme to learn a CANN from real images. Firstly, mimicking the dorsal visual pathway, we adopt a deep neural network trained by a large number of nature images to generate neural representations of objects. It has been observed that the neural representations generated by a deep neural network (i.e., the activities in the representation layer preceding the reading-out layer) can explain a large portion of statistical characteristics of the real neural data recorded in the higher visual cortexes [17], indicating that the representations generated by deep learning capture some sematic information of objects. Secondly, mimicking the hippocampus, we adopt a modified Hebb rule to construct neuronal connections based on the representations of a deep neural network. This modified Hebb rule encodes the correlations between objects' representations into the structure of the network. We carry out experiments, which demonstrate that if the presented images are linked by a continuous feature, a CANN is learned successfully.

2 Learning Correlated Patterns

To start, we introduce how to store correlated patterns in a recurrent network, presuming that the proper neural representations of objects have been generated already (to be discussed in Sec.4). It is well known that the conventional Hebb rule can only learn statistically uncorrelated patterns, resulting in the classical Hopfield model[11]. For correlated patterns, their associated stationary states will merge into only a few attractor states. To solve this issue, several strategies have been proposed. Specifically, Kroff and Treves[1] proposed a popularity-based approach, whose idea is reduce the learning ratio of popular neurons (i.e., those neurons which are involved in many memory patterns). This method works well only under the condition that neuronal activities in a memory pattern are statistically independent, which is hard to be satisfied in practice (Fig.1). Blumenfeld et al.[2] proposed a novelty-based method, which dynamically adjusts the learning ratio relying on the difference between the newly arriving pattern and those already stored in the network. This novelty-based learning idea explains several interesting neurobiological experiments [3, 4]. However, the way of modifying the learn ratio in this approach is empirical and can not guarantee that it works well always (Fig.1).

Here, we adopt an orthogonal Hebb rule to learn correlated patterns [5]. This method works well and is also biologically plausible. Its basic idea is that when a pattern ξ^{p+1} is presented, the network first orthogonalizes it with respect to the stored patterns, and then learns the orthogonalized pattern η^{p+1}. The latter is calculated to be,

$$\eta^{p+1} = \xi^{p+1} - \sum_{\mu=1}^{p} \hat{\eta}^{\mu} \hat{\eta}^{\mu} \xi^{p+1}, \tag{1}$$

where $\hat{\eta}^{\mu} = \eta^{\mu}/|\eta^{\mu}|$ is the normalization of η^{μ}.

According to the Hebb rule, the connection strength between neurons i and j is set to be,

$$W_{ij} = \sum_{\mu=1}^{P} (\hat{\eta}_i^{\mu} \hat{\eta}_j^{\mu} - \delta_{ij} \hat{\eta}_i^{\mu} \hat{\eta}_i^{\mu}) \tag{2}$$

where P is the total number of patterns, and δ_{ij} is Kronecker delta function, i.e., $\delta_{ij} = 1$ for $i = j$ and otherwise $\delta_{ij} = 0$.

Denote $\mathbf{S} = \{S_i\}$, for $i = 1, 2, ..., N$, to be the network state, with S_i taking values of ± 1. The network dynamics is given by

$$S_i(t+1) = sign(\sum_j W_{ij} S_j), \tag{3}$$

and the energy function of the network state is calculated to be

$$E = -\frac{1}{2} \mathbf{S}^T \mathbf{W} \mathbf{S}. \tag{4}$$

Note that the network dynamics contains a novelty-detection process. Suppose a pattern ξ^{P+1} is already learned as the pattern ξ^v, for $v < P+1$, in the network. When this pattern is presented, we have

$$\begin{aligned}
\mathbf{W}\xi^{p+1} &= \mathbf{W}\xi^v, \\
&= \sum_{\mu=1}^{v-1} \hat{\eta}^{\mu} \hat{\eta}^{\mu,T} \xi^v + \hat{\eta}^v \hat{\eta}^{v,T} \xi^v + \sum_{\mu=v+1}^{P} \hat{\eta}^{\mu} \hat{\eta}^{\mu,T} \xi^v - O(\frac{P}{N}) \xi^v, \\
&= \xi^v - \eta^v + \eta^v - O(\frac{P}{N}) \xi^v, \\
&= [1 - O(\frac{P}{N})] \xi^v.
\end{aligned} \tag{5}$$

Thus, according to Eq.(3), this pattern is recognized by the network as learned previously.

Although the above orthogonal Hebb rule may appear to be quite mathematic, its fundamental idea can be implemented in the neural system. For instances, the orthogonal operation may be implemented in the dentate gyrus of the hippocampus, where pattern separation is known to be performed, the novelty detection may be implemented in CA3 [9, 10], and the normalization operation is the canonical neural computation [8]. Fig.1 shows the results of applying different methods to learn the highly correlated handwriting numbers. We see that only the orthogonal Hebb rule works well.

Fig. 1. Retrieval performances of different learning rules. From top to bottom: the original ten digital numbers, the result of the conventional Hebb rule, the result of the popularity-based method, the result of novelty-based method, and the result of the orthogonal Hebb rule.

3 Learning a CANN from Continuous Morphed Patterns

3.1 A CANN of discrete dynamics

Before learning real images, we first try synthetic morphed patterns. As illustrated in Fig.2A, these morphed patterns are constructed by shifting an active patch (taking values of ones, and all other units are zeros) step by step along the network. We select a set of continuously morphed patterns as the memories to be stored in the network, and use the orthogonal Hebb rule to construct the neuronal recurrent connections. The network dynamics follows Eq.(3). After learning, the energies of the network states given by Eq.(4) are shown in Fig.2B. We see that those continuously morphed patterns are successfully learned as attractors (local minima) of the network. Furthermore, these attractors form a relatively flat valley of low energy in the network state space, and with more morphed patterns stored, the valley becomes more flat (see the inset in Fig.2B). We also find that the learned neuronal recurrent connections are shift-invariant along the morphing parameter (Fig.2C), a key property of CANNs. Overall, the conclusion is that when continuously morphed patterns are presented, a CANN is learned by the orthogonal Hebb rule.

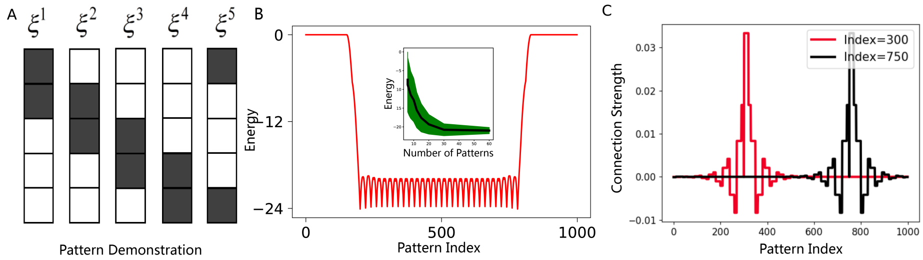

Fig. 2. (A) Illustrating the construction of morphed patterns. The network size $N = 5$ and the patch size $L = 2$. (B) The energies of all morphed patterns in the learned CANN. The network size $N = 1000$. Totally 1000 morphed patterns are constructed, with the patch size $L = 50$, and the patterns are indexed from 1 to 1000. We choose those patterns indexed from 200 to 800 to be the memory patterns. The inset shows the relationship between the energy of the valley and the number of stored patterns. The black line represents the mean of the valley energy averaged over different realizations of selecting stored patterns, and the green shadow denotes the standard deviation. (C) The learned recurrent connections from two neurons at different locations, which are translation-invariant.

3.2 A CANN of continuous dynamics

To be biologically more plausible, we extend the above discrete model to be a continuous one, which is written as,

$$\tau \frac{dV_i}{dt} = -V_i + \sum_j W_{ij} g(V_i) + I_i, \tag{6}$$

where V_i denotes the state of the ith neuron, τ the time constant, I_i the external input, and $g(V_i) = \frac{1}{1+e^{-\alpha * V_i}}$ is a nonlinear function.

Consider that the network learns a set of continuously morphed (shift) gaussian bumps (Fig.3A). We construct the neuronal connections $\mathbf{W}$ according to the orthogonal Hebb rule, which is translation-invariant among neurons (Fig.3B). Fig.3C shows the attractors held by the network after learning, which form a one-dimensional manifold embedded in the high-dimensional space, i.e., a CANN is learned. Note that there are distortions between the training patterns and the attractors held by the network (Fig.3A), nevertheless, one-to-one correspondences between them are established (Fig.3C), meaning that the information of training patterns is acquired by the network. We can also diminish these distortions by applying the stationary states of the network as training patterns iteratively.

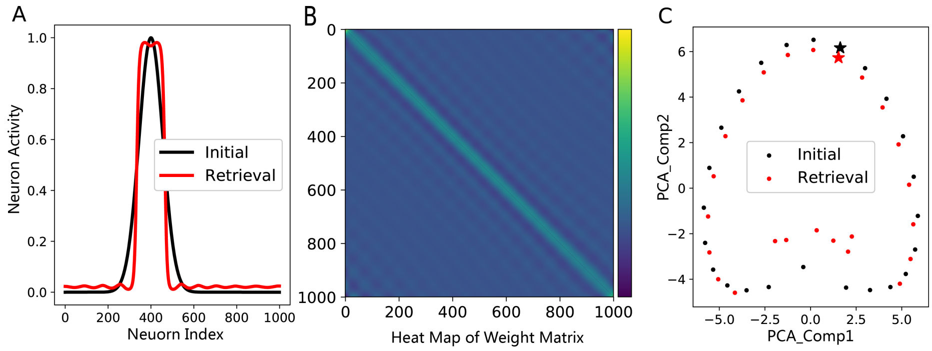

Fig. 3. (A) A training gaussian-shaped pattern (black) and the corresponding attractor learned by the network (red). (B) The translation-invariant recurrent connections between neurons after a set of continuously morphed gaussian bumps are learned. (C) The one-to-one correspondences between the training patterns and the attractors learned by the network. Using the first two dominant PCA components, it is shown that the attractors of the network form a one-dimensional manifold. The parameters: $\tau = 1, \alpha = 8$.

4 Learning a CANN from Real Images

4.1 Generating suitable neural representations

In the above study, we have presumed that the proper neural representations of objects are already available. In practice, this is not a trivial issue. Essentially, an unsupervised Hebb rule is to convert the correlation (overlap) between neural representations into the connection form between neurons, such that the network encodes not only objects but also their relationships. Thus, it is crucial that the neural representations of objects are organized, in term of that their correlation reflects a sematic, rather than a superficial, relationship between objects. In other words, we expect that the closeness between two neural representations should reflect that the two objects they represent are similar in categorization, rather than that the two objects are close in spatial locations. If the raw images of objects are used as the neural representations, then the network will not learn a meaningful result: the network will treat the same object at different locations to be different, whereas treat different objects at the same location to be similar. So, how does the neural system generate proper representations of objects?

Recently, the model of deep neural networks, which mainly mimics the feedforward, hierarchical information processing in the dorsal visual pathway, has made great success in object recognition. Interestingly, it has been observed that in spite of the simple structure, a deep neural network pre-trained by a large number of natural images generates object representations (the neural activity preceding the read-out layer), which capture some statistical characteristics of neural responses in the higher visual cortex. It is known that the neural representations in the higher visual cortex are much more abstract than raw images and those in the early visual cortex. Therefore, a deep neural network provides a way to generate neural representations having certain-level of sematic meaning. In the below, we adopt a deep neural network (DNN) adapted from the Oxford VGG CNN-S network [18] trained by ImageNet to generate neural representations for objects.

4.2 Learning a CANN from continuously rotating chairs

We collected a dataset of rotating chairs in an indoor environment. The telescopic chair was taken by a digital camera with a white wall background, and the chair was rotated from 0^o to 360^o. Totally, 679 images of the chair were obtained, and they were cropped into the size of 224×224 to fit the input of the DNN. No other preprocessing to the inputs was done except normalization to $[0, 1]$. The neural representations generated by the DNN were then used as the memory patterns to construct a CANN. Fig.4A shows that indeed a CANN is learned, and the representations of the rotating chairs form a subspace of attractors.

To further confirm that the network does have the CANN structure, we carry out the mental rotation experiment. Mental rotation is an important characteristic of CANNs, an property coming from the flat subspace of low-energy in a CANN, and was observed in the experiment [19]. Mimicking the experimental setting, we set the network state to be stable initially at an angle, and then apply an external input pointing to a target angle. Under the drive of the external input, the network state changes from the initial to the target position. We record how the network state evolves during this process by measuring the instant overlap between the network state and all attractor states. Interestingly, we observe that if the target angle is not far away from the initial angle, the network state (a bump) rotates smoothly from the initial angle to the target one, displaying the mental rotation behavior as observed in the experiment (Fig.4B).

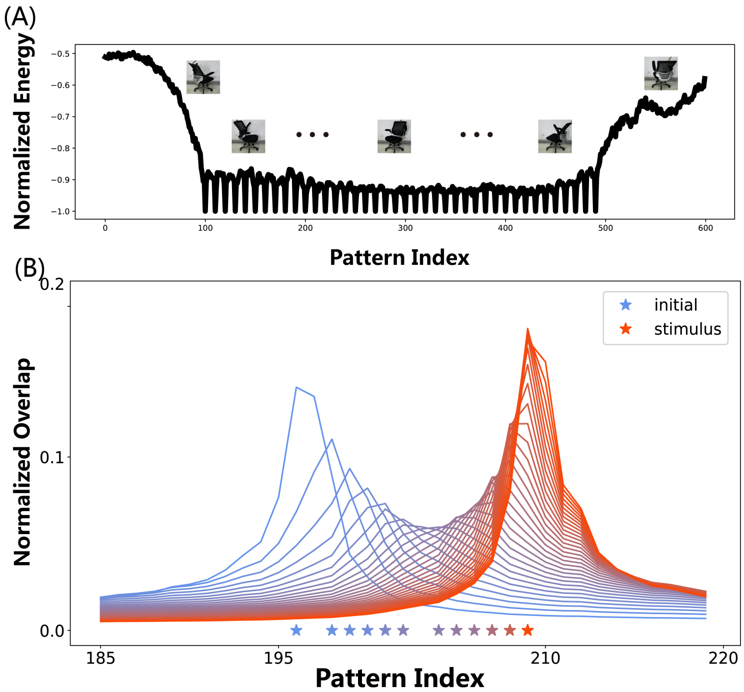

Fig. 4. (A) A CANN learned from continuously rotating chairs. The attractor corresponding to different rotating angles form a subspace of low energy. (B) Mental rotation in the CANN. Starting from an initial angle, the network state (bump) rotates smoothly to the target angle where the stimulus is applied. The parameters are the same as in Fig.3.

5 Conclusion

The canonical neural representation model of CANNs is receiving increasing attention in both neuroscience and brain-inspired computation. In the present study, we explore the long-standing, unresolved question in the field, i.e., how the neural system acquires the CANN structure in practice. We focus on solving two technical challenges, namely, how to generate neural representations of sematic meaning and how to store correlated patterns in a neural network. For the former, we consider that a deep neural network, which mimics the dorsal visual pathway, generates suitable neural representations. For the latter, we show that the orthogonal Hebb rule, which may be achieved in the hippocampus, can learn correlated patterns efficiently. Tested on synthetic data and real images of rotating chairs, we demonstrate that our method learns CANNs successfully. We also test others real images and confirm that our method works very well (data not shown).

References

1. Kropff E, Treves A. Uninformative memories will prevail: the storage of correlated representations and its consequences[J]. HFSP journal, 2007, 1(4): 249-262.
2. Blumenfeld B, Preminger S, Sagi D, et al. Dynamics of memory representations in networks with novelty-facilitated synaptic plasticity[J]. Neuron, 2006, 52(2): 383-394.
3. Leutgeb J K, Leutgeb S, Treves A, et al. Progressive transformation of hippocampal neuronal representations in "morphed" environments[J]. Neuron, 2005, 48(2): 345-358.
4. Wills T J, Lever C, Cacucci F, et al. Attractor dynamics in the hippocampal representation of the local environment[J]. Science, 2005, 308(5723): 873-876.
5. Srivastava V, Sampath S, Parker D J. Overcoming Catastrophic Interference in Connectionist Networks Using Gram-Schmidt Orthogonalization[J]. PloS one, 2014, 9(9): e105619.
6. Kim, Sung Soo, et al. Ring attractor dynamics in the Drosophila central brain. Science 356.6340 ,2017: 849-853.
7. Wu S, Wong K Y M, Fung C C A, et al. Continuous attractor neural networks: candidate of a canonical model for neural information representation[J]. F1000Research, 2016, 5.
8. Carandini M, Heeger D J. Normalization as a canonical neural computation[J]. Nature Reviews Neuroscience, 2012, 13(1): 51-62.
9. Leutgeb J K, Leutgeb S, Moser M B, et al. Pattern separation in the dentate gyrus and CA3 of the hippocampus[J]. science, 2007, 315(5814): 961-966.
10. Yassa M A, Stark C E L. Pattern separation in the hippocampus[J]. Trends in neurosciences, 2011, 34(10): 515-525.
11. Hopfield J J. Neural networks and physical systems with emergent collective computational abilities[J]. Proceedings of the national academy of sciences, 1982, 79(8): 2554-2558.
12. Seelig J D, Jayaraman V. Neural dynamics for landmark orientation and angular path integration[J]. Nature, 2015, 521(7551): 186-191.
13. Amari S. Dynamics of pattern formation in lateral-inhibition type neural fields[J]. Biological cybernetics, 1977, 27(2): 77-87.
14. Zhang K. Representation of spatial orientation by the intrinsic dynamics of the head-direction cell ensemble: a theory[J]. Journal of Neuroscience, 1996, 16(6): 2112-2126.
15. Logothetis N K, Pauls J, Bülthoff H H, et al. View-dependent object recognition by monkeys[J]. Current biology, 1994, 4(5): 401-414.
16. Mante V, Sussillo D, Shenoy K V, et al. Context-dependent computation by recurrent dynamics in prefrontal cortex[J]. Nature, 2013, 503(7474): 78-84.
17. Yamins D L K, Hong H, Cadieu C F, et al. Performance-optimized hierarchical models predict neural responses in higher visual cortex[J]. Proceedings of the National Academy of Sciences, 2014, 111(23): 8619-8624.
18. Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition[J]. arXiv preprint arXiv:1409.1556, 2014.
19. Georgopoulos, Apostolos P., Masato Taira, and Alexander Lukashin. "Cognitive neurophysiology of the motor cortex." Science 260.5104 (1993): 47-52.